

Early-life exposure to weather shocks and learning outcomes in India*

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Abstract

We study the medium-term effects of early-life exposure to weather shocks on children's learning outcomes in India. Using spatial and cohort variation in exposure to a plausibly exogenous flood within difference-in-differences and event study frameworks, we find that children exposed to an unanticipated and extreme flood in early life score 0.08 and 0.06 standard deviations lower in reading and math tests, respectively, five to six years after exposure. Further, using an event study design, we find that the timing of exposure matters, with significant negative effects on cognitive development observed only for exposure just prior to school-start age (i.e., ages four and five), while exposure at younger ages shows no significant impact. The results are driven by children from lower socioeconomic backgrounds, suggesting sudden and persistent income shock as an underlying mechanism. Our findings provide fresh evidence on the extent to which the timing of exposure matters and have important policy implications.

Keywords: *Learning Outcomes; Education; Flood; Kosi; India*

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1. Introduction

An extensive literature has documented that shocks during early life (in utero and zero to five years) can have long-term consequences in later childhood or adulthood.¹ A major argument in this literature is that the timing of exposure matters with the effect that the same shock can have different effects at different ages at which the child is exposed (Webb, 2024). Which stages of early life are critical and what groups of children should policy interventions focus on is an open empirical question. A large chunk of this literature suggests that shocks in utero and during the early phase of life (zero to two years) are extremely sensitive to childhood development (Currie & Almond, 2011). However, the empirical evidence on the extent to which the timing of exposure matters is quite mixed, with studies finding different results in different contexts.²

In this paper, we study the persistent effects of early-life exposure to an extreme and unanticipated flood on learning outcomes in childhood and explore whether the timing of exposure matters. The context of our paper is the Indian state of Bihar — which is one of the poorest regions in the world. In August 2008, it experienced an unprecedented flood in the *Kosi* River (first in last 50 years), which was dubbed a catastrophe (UNDP India, 2009) and led to heavy economic losses. About 1,000 villages in five districts of eastern Bihar were severely impacted by the flood, with two additional districts being moderately affected. We present a brief background in the following section.

We use nationally representative data (at the district level) on language and math test scores of children aged 5-16 in 2013 and 2014 (i.e., five to six years after the flood) from the Annual Status of Education Report (ASER) surveys. Our identification strategy exploits two sources of variation. First, we exploit cohort variation in exposure to the flood — the sampled children are born between 1997 and 2009. Our treatment group includes children who were in utero or aged zero to five during the flood in 2008. The control group includes children aged six or above in 2008. Second, we exploit spatial variation in exposure to the flood as described above. We compare treated and control children in affected districts with unaffected districts under a difference-in-differences framework.

We find that children exposed to the flood in utero or during ages zero to five perform worse on basic language (grade one or two equivalent) and math (grade two or three equivalent) tests five to six years after the exposure. Specifically, reading and math scores deteriorate by about 0.08 and 0.06 standard deviations, respectively. These difference-in-differences estimates capture the combined effects of being exposed to the shock between in

¹For a comprehensive review of literature, please see Almond et al. (2018).

²For example, while early-life extreme rainfall shock in Mexico is found to adversely impact cognitive outcomes later on only for exposure during in utero (Aguilar & Vicarelli, 2022), the same shock is found to adversely impact cognitive outcomes for exposure during ages 0-2 but not in utero in Ecuador (Rosales-Rueda, 2018). On the contrary, weather shocks in Indonesia positively affect cognitive outcomes for exposure only at age 2 (Webb, 2024).

utero and age five. However, using event study design, we identify that exposure just prior to the school-start age (i.e., at ages four and five) drives the main results and, unlike most of the existing empirical studies on weather shocks, we do not find adverse effects for children exposed to the flood in utero or before age two. Further, our heterogeneity analysis shows that results are driven by children from lower socioeconomic backgrounds. As a potential mechanism, we do not find evidence of early-life nutritional deprivation driving these results. However, using night-time lights data, we find huge and persistent economic losses associated with the flood. We perform a series of robustness checks to further validate these results.

This paper is related to a much broader literature studying the effects of early-life exposure to adverse shocks on various economic outcomes. A wealth of literature has focused on investigating the effects of adverse shocks in early life, including environmental shocks, epidemics, nutrition shocks, maternal stress, weather anomalies and conflicts on a range of outcomes. For instance, early-life exposure to adverse shocks has been found to adversely affect long-term health (Aizer et al., 2016; Lu et al., 2023; Groppo & Kraehnert, 2016; Andalón et al., 2016), education (Adhvaryu et al., 2024; Shah & Steinberg, 2017; Almond, 2006) and employment and earnings (Karbownik & Wray, 2019; Scholte et al., 2015), among other such outcomes (see reviews by Almond et al. (2018) and Currie & Almond (2011) and the references therein).

However, we primarily contribute to a small but growing literature studying the effects of early-life exposure to weather shocks on cognitive outcomes (Webb, 2024; Rosales-Rueda, 2018; Aguilar & Vicarelli, 2022; Pazos et al., 2024; Shah & Steinberg, 2017; Leight et al., 2015). Broadly, these studies find that adverse weather shocks negatively affect the cognitive outcomes of children and the shocks during the in utero period or prior to age two are particularly more relevant for later-life outcomes (Pazos et al., 2024; Rosales-Rueda, 2018; Aguilar & Vicarelli, 2022; Webb, 2024). Our contributions to this literature are threefold.

First, we contribute by providing fresh evidence on the timing of exposure during early life, at which floods adversely affect cognitive outcomes in later childhood. To the best of our knowledge, this is the first paper to document that early-life exposure to weather shocks adversely affects the cognitive development of children exposed just at the school-start age (years four and five). Notably, we do not find negative effects of exposure to flood in utero or prior to age two, which contrasts with much of the prior evidence on the timing of exposure and underscores the importance of examining broader childhood developmental windows. These findings are important because they highlight that the extent to which timing of early-life shocks matters for cognitive development later in childhood often depends on the underlying mechanisms and the socioeconomic and institutional contexts — such as household access to consumption-smoothing mechanisms, public safety nets and the availability of post-disaster recovery strategies.

Second, this paper is among the very few to study the medium-term effects of early-life exposure to floods on cognitive outcomes. We believe documenting these intermediate effects is crucial from a policy perspective since the emerging trends in global climate change suggest an increase in both the frequency and unpredictability of weather shocks (Cai et al., 2014). Given that households in developing countries are susceptible to weather shocks and childhood learning outcomes could predict later-life productivity (Heckman et al., 2006), documenting these medium-term effects is crucial for designing effective post-disaster policies to mitigate long-term consequences.

Third, our paper differs significantly from the existing papers cited above in terms of research design and data. It leverages a plausibly exogenous flood shock and rich data to better understand the importance of timing of early-life exposure to weather shocks. Unlike previous studies that heavily rely on rainfall shocks, we exploit a plausibly exogenous, extreme and unanticipated flood shock that resulted from an embankment breach (originating from Nepal), generating within-state variation in flood exposure across birth cohorts and reducing potential biases due to confounding factors.³ Further, these studies compare cohorts born during and after the rainfall shocks, assuming that not-yet-born children serve as a valid counterfactual group.⁴ However, if the adverse income shock associated with precipitation persists over time, later-born children may also be affected, weakening identification. By contrast, our paper uses an exogenous flood shock and a unique dataset that provides us with a more robust control group — children who were already in school and old enough at the time of the flood. Since our test scores measure basic literacy and numeracy skills irrespective of age, these relatively older children would have mostly acquired these skills and are much less likely to be affected by the flood. This approach strengthens the identification and provides new insights into the timing of exposure and long-term human capital accumulation.

The rest of this paper proceeds as follows: Section 2 presents a brief background. Section 3 describes data. Section 4 lays out the empirical strategy. Section 5 presents results. Section 6 provides a brief discussion. Section 7 concludes.

³Confounding factors such as a productivity shock associated with rainfall shocks could adversely impact cognitive outcomes for exposure during different ages. In India, for example, exposure between in utero and age two improves learning outcomes, but exposure at older ages (five and above) worsens cognitive outcomes (Shah & Steinberg, 2017). In Indonesia, exposure to weather shocks improves cognitive outcomes for adult males who were exposed to these shocks at age two, with exposure at other ages having no effects (Webb, 2024).

⁴For instance, Rosales-Rueda (2018) compares later-born cohorts with cohorts born during rainfall shock to find that in utero exposure to floods translates to worse health and cognitive outcomes in Ecuador. Aguilar & Vicarelli (2022) also uses a similar specification to estimate the effects of early-life rainfall shocks in Mexico.

2. Background

The *Kosi* flood, which occurred on August 18, 2008, had catastrophic consequences for many areas in the northeastern region of Bihar. The heavy monsoon rains caused a breach in the embankment near the village of *Kusaha* in neighboring Nepal, redirecting the *Kosi* River to an ancient course it had abandoned around 200 years ago (Khanna & Kochhar, 2023). This event represents the most severe flood in Bihar’s recent history. Unlike earlier floods in the state, this 2008 flood was unprecedented and extraordinary, as the river burst through embankments, altering its course and inundating areas that had remained flood-free for over 50 years (Dixit, 2009). This sudden and unexpected breach left many parts of northeastern Bihar completely submerged in water, affecting about 1,000 villages across five severely impacted and two moderately affected districts (see Figure 1 for details of affected districts). Since the population in these districts had not experienced floods for over 50 years, the massive *Kosi* flood of 2008 caught residents completely unprepared, affecting nearly three million people in the region and requiring immediate evacuation of one million from the submerged areas (UNDP India, 2009).

[Figure 1 about here]

The *Kosi* flood is reported to have severely disrupted the local economy and livelihoods, causing extensive damage at both the village and household levels. A survey conducted by the UNDP India (2009) revealed that approximately about 93% households in the affected regions of Bihar experienced income losses exceeding 50% during the first three months following the disaster in August 2008. These economic losses represented a significant dent in household incomes, particularly as the majority of the affected population belonged to economically modest backgrounds, heavily reliant on casual labor and agricultural productivity to sustain their daily needs.

The flood also inflicted enormous economic losses to the state, estimated at approximately USD 135 million, which accounted for 0.5% of Bihar’s GDP in 2008 (Khanna & Kochhar, 2023).⁵ Beyond the economic impact, the natural disaster has also been reported to have caused massive damage to infrastructure, including houses, livestock, and crops, directly affecting the livelihoods of around 500,000 farmers in the region.

3. Data

In this paper, we use data from the Annual Status of Education Report (ASER) survey (ASER, 2014, 2015). ASER surveys are household-based surveys that have been conducted in nearly

⁵Khanna & Kochhar (2023) independently verify using night lights data that flood led to massive economic losses in the state.

every rural district of India since 2005. The survey is representative at the district level and primarily aims to document the educational status of children aged 3-16 in rural India. The surveys are typically conducted between September and November each year.⁶ The ASER surveys collect data on children within their households, regardless of their enrollment status. For our purpose, we use the 2013 and 2014 rounds of the ASER survey, which were conducted five to six years after the 2008 *Kosi* flood in Bihar. We restrict our analytical sample to children from Bihar to investigate the medium-term effects of early-life exposure to the *Kosi* flood on basic literacy and numeracy skills.

Children aged 5-16 are assessed on basic language (reading) and arithmetic skills, with each test comprising four components (see Appendix Figure A.1 for details). The reading score ranges from 0 to 4, where 0 indicates that the child cannot read anything, 1 indicates the ability to read letters, 2 indicates the ability to read words, 3 indicates the ability to read a paragraph, and 4 corresponds to the ability to read a story (equivalent to grade two). Similarly, the math score ranges from 0 to 4, where 0 indicates no proficiency, 1 indicates recognition of single-digit numbers, 2 indicates recognition of double-digit numbers, 3 indicates the ability to solve subtraction problems (equivalent to grade two), and 4 corresponds to the ability to perform division problems (equivalent to grade two or three). In each test, the children are evaluated at the highest level they achieve, meaning that if a child can read a story, they can also read at the lower levels; however, if they can only read words, they are unable to read a paragraph or a story. Notably, the same tests are administered to all children, regardless of their age and grade level.

For our analysis, following existing literature (Fagernäs & Pelkonen, 2020; Krämer et al., 2021; Lahoti & Sahoo, 2020), we standardize the reading and math scores so as to interpret our coefficients as standard deviations. Specifically, we standardize these scores with respect to the mean and standard deviation of the control group (i.e., children from unaffected districts), such that the standardized test scores have a mean of 0 and a standard deviation of 1 for these children. Additionally, we create a variable for grade-for-age or the probability of being on track. This variable takes a value of 1 if a child is enrolled in a grade appropriate for their age (with a typical school-start age in India being 6 years). For example, the variable takes a value of 1 for a child who is 8 years old and enrolled in grade 2 or higher and 0 if the child is enrolled in grade 1 or lower.

In addition to test scores, the ASER surveys collect data on basic household characteristics, including maternal age and education, household size and access to electricity and toilets. Although the survey does not collect data on household incomes, it records housing type to measure the socioeconomic status (SES) of the household. Specifically, housing is classified as concrete, semi-concrete or thatch/mud.

For our purpose, we define a child's socioeconomic status as higher (i.e., non-poor) if

⁶However, since 2014, the surveys have been conducted every alternate year.

they reside in a concrete or semi-concrete dwelling and lower (poor) otherwise. In absolute terms, higher SES households tend to have more stable housing and are more likely to have access to basic household amenities such as electricity and toilets. Lower SES often live in temporary structures with limited access to these amenities, reflecting extreme poverty.

[Table 1 about here]

Table 1 presents the summary statistics. The mean reading and math score is about 2. Given the ordinal nature of these numbers, it means that, on average, a child can read words but not a paragraph or a story. Moreover, she can recognize two-digit numbers but cannot solve subtraction or division problems. The mean and standard deviation of the standardized scores are not exactly zero and one, respectively, because we standardize with respect to children from unaffected districts.⁷ While all children in our sample are currently enrolled, about 81% are enrolled in grades appropriate for their age. An average child is around 10 years old and is currently enrolled in class five. About 50% of the children in our sample are boys.

4. Empirical Strategy

Our identification strategy exploits spatial and cohort variation in exposure to unexpected and extreme flooding in the Indian state of Bihar under a difference-in-differences framework.⁸ We compare children exposed to the flood in their early life (in utero and aged zero to five) in flood-affected districts with unaffected districts. Our control group consists of children aged six to eleven at the time of the flood in 2008. We estimate the following specification:

$$y_{ibdt} = \alpha + \beta(Treated_d * Post_{ib}) + X_{ibdt}\theta + \delta_d + \gamma_b + \tau_t + \phi_{dt} + \epsilon_{ibdt} \quad (1)$$

y_{ibdt} is the standardized test score outcome of child i born in year b surveyed in district d in year t . $Treated_d$ is an indicator that takes the value 1 if the child resides in flood-affected district d and 0 otherwise. $Post_{ib}$ is defined as a dummy taking value 1 for child i being in utero or aged zero to five at the time of flood and 0 otherwise. X_{ibdt} is a vector of child and household-level characteristics a child's gender and grade, her mother's age and schooling status, socioeconomic status, household size, electricity and toilet availability. δ_d , γ_b , τ_t and ϕ_{dt} respectively denote vectors of district, birth cohort, survey year and district-survey year fixed effects. District fixed effects control for time-invariant district-level characteristics. Cohort and survey year fixed effects, respectively, control for shocks common to all children

⁷While the standardized scores are slightly skewed to the left because of average scores being closer to the highest value (4) than to the lower value (0), this is unlikely to bias our results.

⁸Karbownik & Wray (2019) use a similar specification to estimate the long-term effects of natural disasters in the US.

born in the same year and surveyed in the same year. District-survey year fixed effects allow test score variables to evolve non-linearly over time. The standard errors are clustered at the district level.

The identification in this framework crucially hinges on the parallel trends assumption, which, in our case, implies that the test score outcomes of older cohorts in affected districts would have evolved in parallel to those of unaffected districts. To test the validity of parallel trends assumption and to pin down specific ages at which exposure to flood affects later-life test scores, we use the following event study design:

$$y_{ibdt} = \alpha + \sum_{b=1997(b \neq 2002)}^{2009} \beta_b \mathbf{1}_{ib} * Treated_d + X_{ibdt}\theta + \delta_d + \gamma_b + \tau_t + \phi_{dt} + \epsilon_{ibdt} \quad (2)$$

Here, the dummy variable $\mathbf{1}_{ib}$ takes the value 1 if the child i is born in year b and zero otherwise. The rest of the specification is the same as equation 1. Children born in the year 2002 (i.e., six years old at the time of the flood in 2008) serve as a reference group. Coefficients β_b 's capture differences in test scores of children affected and unaffected by birth cohorts.

5. Results

We present results in Table 2. In columns 1 to 3 and 4 to 6, the dependent variables are standardized reading and math scores, respectively (the *z-scores*). Column 1 shows that early-life exposure to the flood reduces the reading score by about 0.08 standard deviations six years after the exposure. The negative effect is statistically significant at the 1% level. Column 1 includes the district and cohort fixed effects and a bunch of control variables for child and household characteristics such as the child's gender and grade, her mother's age and schooling status, socioeconomic status, household size, electricity and toilet availability.⁹ In columns 2 and 3, we respectively introduce the survey year fixed effects and the district-survey year fixed effects. The estimated coefficients remain magnitudinally stable and retain statistical significance at 1% level upon the inclusion of these additional fixed effects.

Column 4, which only includes the district and cohort fixed effects and a bunch of control variables for child and household-level characteristics, shows that exposure to the flood early in life negatively affects math test scores five to six years later. In terms of magnitude, the math score deteriorates by about 0.06 standard deviations, and the effect is statistically significant at the 5% level. However, the negative effect is robust to the inclusion of the survey year fixed effects (Column 5) and the district-survey year fixed effects (Column 6). The estimated coefficients remain magnitudinally stable and retain statistical significance at

⁹In Appendix Table A.1, we show that these results are also robust to the exclusion of district fixed effects and cohort fixed effects.

the 5% level.

[Table 2 about here]

Next, we present event study results from estimating equation 2 to validate the parallel trends assumptions and to pin down specific ages at which exposure to the flood has an effect on test scores five to six years after the exposure. Figure 2 presents the results. It reports the coefficient estimates and associated 95% confidence intervals. Panels (a) and (b) show event study graphs for language and math test scores, respectively. Children born in the year 2002 (aged six at the time of the flood in 2008) serve as the reference group. We note two key observations from event study estimates in both panels of Figure 2. First, the gaps in the test scores of children aged seven or older in the year 2008 in flood-affected and unaffected districts of Bihar are close to zero and statistically insignificant. These results validate our identifying assumption of parallel trends.

Second, we observe a sharp decline in the test scores of children aged four and five at the time of the flood in affected districts vis-à-vis their counterparts in unaffected districts. For these two cohorts, the negative effects are as large as 0.11 and 0.09 standard deviations, respectively, for the reading and math scores. Further, contrary to existing empirical evidence on the timing of exposure to weather shocks (Webb, 2024; Pazos et al., 2024; Aguilar & Vicarelli, 2022; Rosales-Rueda, 2018), we observe that exposure to the flood during in utero period or prior to age two has no statistically significant effect on test scores in the medium-run. While we also find adverse effects for age two,¹⁰ we find larger effects at ages four and five. These findings suggest that the extent to which the timing of exposure to weather shocks matters depends on the underlying mechanisms as well as the context. More precisely, the existing studies on the timing of exposure find that negative shocks during early life translate into worse outcomes later in childhood, majorly via the nutrition channel. In particular, the adverse nutrition shock could interact with brain development during the early phase of life (in utero and up to two years) and lead to adverse cognitive outcomes later on. However, in our case, the nutritional channel appears unlikely to be driving our results as we do not find negative effects for exposure prior to age two. We investigate the potential underlying mechanisms in the following subsection. We present additional results in the Appendix. Table A.2 shows that the flood has no statistically significant effect on the probability of being on track and the uptake of private tuition.¹¹

We also investigate whether the differences in test scores of children exposed to extreme flood shock appear immediately or only emerge over time.^{12,13} To this end, we conduct a

¹⁰At 10% level for reading score and 5% level for math score.

¹¹Figure A.2 shows event study estimates for these variables.

¹²We thank an anonymous reviewer for pointing this out.

¹³Unfortunately, the available data does not allow for an analysis of long-term impacts to investigate whether these effects persist over time. The ASER surveys include test scores only for children aged 5-16, and the 2018 survey covers children who were all born after 2002. As a result, we lack a suitable comparison group of

parallel analysis of the short-term effects of this shock using the ASER survey data from 2009 and 2010, focusing on children surveyed closer to the flood shock of 2008. We estimate equations 1 and 2 using this dataset and present results in Table A.3 and Figure A.3 in the Appendix. The results indicate no significant short-term effects on reading and math test scores. Instead, the differences in the test scores appear to gradually develop over time, with significant effects becoming evident in the medium term, five to six years after the exposure to the shock. These findings highlight that the impact of flood accumulates over time, potentially via mechanisms such as persistent exposure to income shock induced by 'Kosi' floods.

[Figure 2 about here]

5.1. Heterogeneity Analysis

We explore heterogeneity in treatment effects by gender, socioeconomic status and mother's literacy in Table 3. We start by estimating equation 1 separately for different sub-samples and then use a triple difference (DDD) specification to check whether differential effects are statistically significant. Panel A shows that both boys and girls are adversely affected, and the differential effects are not statistically significant (columns 3 and 6). Panel B shows that the results are driven by children from low socioeconomic backgrounds. However, the coefficient on the triple interaction term is positive but not statistically significant. Finally, Panel C shows that the negative effects of the flood are larger in magnitude for children whose mothers are literate. The differential effects are negative and statistically significant for language tests. Together, these heterogeneity results suggest that potential mechanisms behind the deterioration in test scores may be a sudden and persistent income shock as a result of the extreme Kosi flood (UNDP India, 2009; Khanna & Kochhar, 2023) and reduction in parental investment in children (in terms of time spent on supervision at home), potentially due to reconstruction efforts after such a disaster.

[Table 3 about here]

5.2. Potential Mechanisms

In this subsection, we take insights from our heterogeneity analysis to explore potential underlying mechanisms that explain our findings. First, if nutritional deficiencies were driving our results, we would expect larger effects for boys than for girls, given that they are more susceptible to adverse nutrition shocks, particularly in utero or during infancy (Eriksson et al., 2010; Kraemer, 2000). However, we find no differential effects by gender, nor do

children who were 6 or older in 2008 and surveyed in 2018. This remains one of the limitations of the paper, and future research may evaluate these long-term impacts if other data becomes available over time.

we observe adverse effects for children exposed in utero or before age two, suggesting that nutrition is unlikely to be a leading channel. Nonetheless, to further assess the role of nutrition, we use data from the second wave of the Indian Human Development Survey (IHDS-2, conducted in 2011-12) and estimate the effect of *Kosi* flood on a child’s height-for-age (*z-score*) later in the childhood.¹⁴ We estimate a specification similar to our baseline model (1) and present results in Table 4. We do not find a statistically significant effect of the flood shock on the health status of children three to four years after the exposure. Moreover, the event study estimates in Appendix Figure A.4 further show that the parallel trends assumption holds in the data, with the timing of exposure having no significant adverse effects at any age. While the smaller sample size could potentially lead to imprecisely estimated coefficients, the consistently positive sign of the estimates alleviates concerns that the null findings are solely driven by sample size issues. Therefore, unlike many existing studies, we do not find evidence that nutritional deprivation is a leading channel driving our main results

[Table 4 about here]

Second, while our heterogeneity results show that the effects are driven solely by children from lower socioeconomic backgrounds, we investigate whether income shock associated with the flood of 2008 is the primary channel driving our results. As highlighted in section 2, anecdotal evidence suggests that the flood heavily disrupted the local economy, generating a huge exogenous income shock to the households. We use night-time lights data to independently verify that income shock is the leading driver of our results. To this end, we extract data on average night-time lights from 2004 to 2013 from the SHRUG database (Henderson et al., 2011; Asher et al., 2021) and employ a synthetic control method to assess the effect of the *Kosi* flood on economic activity in the state of Bihar. We present these results in Figure 3. We find a large and persistent income shock caused by the 2008 floods. These results are in line with the findings of Khanna & Kochhar (2023), who use the same data to show that the *Kosi* flood generated huge income losses that persisted over time. Moreover, they further show, using consumption data, that household consumption expenditure also declined heavily after the disaster. To summarize, in conjunction with the anecdotal evidence and existing empirical analyses, income shock could be the leading channel driving our main results.

[Figure 3 about here]

5.3. Robustness Checks

To further validate our main findings, we present results from a battery of robustness checks in the Online Appendix. First, we show that our results are robust to the exclusion of less

¹⁴While IHDS is a panel survey, its coverage is not as extensive as that of ASER data. Therefore, we have a relatively smaller sample of children from Bihar from the IHDS-2.

affected districts in Table A.4. Second, we show that our results go through if we drop the border districts from our analysis (Table A.5). Third, following [Rambachan & Roth \(2023\)](#), we present sensitivity analysis to potential violations of parallel trends for our post-treatment estimates in Figures A.5 and A.6. We show that the post-treatment coefficients are robust to linear and non-linear violations of parallel trends. Fourth, to show that our results are not due to a mere statistical chance, we randomize treatment to districts and cohorts 1,000 times. In each iteration, we estimate our baseline specification and collect the estimated effects. We plot the density of these coefficients along with our original coefficient estimates in Figure A.7 to show that our coefficients lie in the tails of these distributions.

Since the reading and math test scores are ordinal in nature, in our main analysis, following existing literature, we standardized these scores for ease of interpretation. However, as a further robustness check, we account for the ordinal nature of these scores and present results using logit models in Appendix Table A.6. We find very similar results. Next, to address potential concerns about outliers in the data,¹⁵ we exclude a small percentage of these observations from the data and re-run our main regression and event study analyses. We present these results in Appendix Table A.7 and Figure A.8. These regressions yield results consistent with our main results with the full sample. Finally, since our analysis primarily relies on the data from Bihar, a potential concern is whether these findings can be generalized to other contexts.¹⁶ Given that ASER data is one of the largest household surveys in India and is nationally representative at the district level, we believe our findings can also be generalized to other regions. Nonetheless, to further address the external validity of our results, we conduct a parallel analysis of a different flood shock in another Indian state (Uttarakhand).¹⁷ We present these results in Appendix Table A.8. These results are in line with our main findings from Bihar. The consistency in findings from a different flood shock further strengthens the conclusions from our findings.

6. Discussion

This paper studies the medium-term effects of early-life exposure to extreme and unanticipated floods on learning outcomes five to six years after exposure in India. The findings from the paper indicate that exposure to weather shocks in utero and between zero and five years have significant adverse effects on both language and arithmetic skills. While the effects do not appear immediately, the differences in test scores develop over time, with

¹⁵For instance, a small percentage of mothers are aged more than 56 years (less than 1%), and a small percentage of households (less than 1%) have more than 20 members. We thank an anonymous reviewer for these observations.

¹⁶We thank an anonymous reviewer for this comment.

¹⁷The state of Uttarakhand witnessed extreme floods in June 2013, which affected almost all the districts, but five of these were severely impacted. We compare children from four of these five districts with children from the neighboring state of Himachal Pradesh, leaving out one affected district sharing a border with the control state. For details, see <https://nidm.gov.in/pdf/pubs/ukd-pl.pdf>.

significantly large effects becoming evident in the medium term (i.e., five to six years after). Further, while our findings are generally consistent with the literature, using event studies, we uncover that the timing of exposure matters, with exposure just prior to the school-start age (four and five years) being more relevant than exposure at younger ages. Unlike most existing studies, we do not find adverse effects in the medium term for children who are exposed in utero or prior to age two. One plausible explanation for these findings is that the impact is not primarily driven by early-life nutritional deprivation (as found in many existing studies). In fact, we do not find the negative effect of flood shock on nutritional outcomes (height-for-age). Prompt response from the government and active relief efforts could plausibly explain the absence of nutritional deficiency. The anecdotal evidence suggests that the government quickly swung into action and provided temporary shelter, food, medicine and clothing to the millions affected by the natural disaster (UNDP India, 2009). This might have enabled the households to smoothen their consumption, and thus, we do not find adverse effects on nutritional outcomes. In fact, the government continued its efforts to assist the affected population by providing food and cash for a longer term.

Another plausible explanation for finding null nutritional effects, particularly for children exposed in utero, could be because of mortality effects that may potentially impact cohort sizes. In other words, if younger children were more likely to die due to flood exposure, it could bias cohort sizes. However, this was a very low mortality event, with only 527 deaths reported (approximately 0.005% of the population), making mortality effects that could bias cohort size very unlikely. The combination of a low mortality rate and prompt government support suggests that the survival of only stronger infants is unlikely to have biased the cohorts. Moreover, even in the presence of such selection effects, they would likely bias the results toward finding more favorable outcomes for children in utero rather than null effects that we document.

Our heterogeneity analysis shows that the effects are driven by children from lower socioeconomic backgrounds, pointing to a sudden and persistent income shock as a plausible underlying mechanism. Using night-time lights data, we document a huge and persistent income shock associated with the 2008 *Kosi* flood. This may explain why children exposed to the weather shock at the ages of four and five were more impacted by the shock than younger children. Since the income shock persisted for several years, those just entering school in 2008 bore the full burden of household adversity, while the younger siblings were less exposed to this income shock. Moreover, these kids could have also suffered due to a reduction in parental investments, such as time spent on supervision at home. To test this mechanism, we need data on time use both before and after the 2008 floods. Unfortunately, the first wave of time-use data in India is from 2019, and data before that is not available. However, using heterogeneity analysis based on the mother's education,¹⁸ we

¹⁸The reduction in time spent on supervision at home should matter more for children with literate mothers and less otherwise.

provided suggestive evidence that a reduction in parental investment could be a plausible mechanism. The lack of solid evidence for this mechanism remains a limitation of the paper.

The ASER data that we use in this paper is one of the largest household surveys (on educational outcomes) in India and covers over 500 districts. It is nationally representative at the district level, ensuring that it captures a broad range of socioeconomic and educational characteristics across India. While our results are based on Bihar only, they can be generalized to other regions and contexts as well. Bihar is also one of the most populous and backward states in India and is also prone to recurrent flooding. Therefore, studying Bihar provides valuable insights into how extreme weather events impact learning in other developing countries where children face similar vulnerabilities. Nonetheless, we present a series of robustness checks to further validate our main results. In particular, we conducted a parallel analysis using a different flood shock in another Indian state to support the validity of our main findings further.

Our findings have very important implications for the policy to mitigate the adverse impacts of adverse weather shocks on educational outcomes. Given that floods and other weather-related shocks are becoming increasingly more frequent and less predictable, governments should prioritize the following to mitigate its effects on educational outcomes. First, governments should implement early intervention programs such as nutritional supplementation to cushion the adverse effects of natural disasters on learning outcomes.¹⁹ Second, governments should expand their financial assistance programs for the most vulnerable households, particularly those coming from lower socioeconomic backgrounds. These could range from temporary cash transfers to fee waivers. This will help them smooth their consumption without needing to pull their young children out to school to join the labor force. Third, policies should exclusively focus on providing early childhood educational support via expanding access to high-quality preschools and early learning programs in disaster-prone areas. This can help children who are entering school right after a disaster to mitigate the adverse consequences of household disruptions. Finally, to mitigate learning losses, policies should be focused on children entering school during or right after the shock, who bear the greatest burden of the household income shock. These could range from providing tuition subsidies, free school meals, or even conditional cash transfers over a couple of years following a major disaster of this magnitude. Together, these interventions can ensure that disruptions in learning are minimized in the aftermath of such huge natural disasters and prevent long-term human capital deficits.

¹⁹In fact, in the present study, we find that the prompt and targeted response from the government to support affected families (in terms of providing food and shelter) helped mitigate adverse effects due to poor nutrition.

7. Conclusion

We study the effect of early-life exposure to flood on learning outcomes in India and investigate the extent to which the timing of exposure matters. We find that exposure to flood during early-life, especially during ages four and five, has a negative effect on learning outcomes five to six years after exposure. Contrary to existing evidence on critical periods of childhood development, we do not find adverse effects for children exposed in utero or before age two. Our findings provide fresh evidence on the extent to which timing of exposure matters and have important policy implications, particularly in developing country contexts like India, which are extremely susceptible to ever-increasing and unpredictable weather shocks in the wake of climate change.

References

- Adhvaryu, A., Molina, T., Nyshadham, A., & Tamayo, J.** (2024). Helping children catch up: Early life shocks and the PROGRESA experiment. *The Economic Journal*, 134(657), 1–22. <https://doi.org/10.1093/ej/uead067>
- Aguilar, A. & Vicarelli, M.** (2022). El Niño and children: Medium-term effects of early-life weather shocks on cognitive and health outcomes. *World Development*, 150, 105690. <https://doi.org/10.1016/j.worlddev.2021.105690>
- Aizer, A., Stroud, L., & Buka, S.** (2016). Maternal stress and child outcomes: Evidence from siblings. *Journal of Human Resources*, 51(3), 523–555. <https://doi.org/10.3368/jhr.51.3.0914-6664R>
- Almond, D.** (2006). Is the 1918 influenza pandemic over? Long-term effects of in utero influenza exposure in the post-1940 US population. *Journal of Political Economy*, 114(4), 672–712. <https://doi.org/10.1086/507154>
- Almond, D., Currie, J., & Duque, V.** (2018). Childhood circumstances and adult outcomes: Act II. *Journal of Economic Literature*, 56(4), 1360–1446. <https://doi.org/10.1257/jel.20171164>
- Andalón, M., Azevedo, J. P., Rodríguez-Castelán, C., Sanfelice, V., & Valderrama-González, D.** (2016). Weather shocks and health at birth in Colombia. *World Development*, 82, 69–82. <https://doi.org/10.1016/j.worlddev.2016.01.015>
- ASER** (2014). *Annual Status of Education Report (Rural) 2013*. ASER Center New Delhi.
- ASER** (2015). *Annual Status of Education Report (Rural) 2014*. ASER Center New Delhi.

- Asher, S., Lunt, T., Matsuura, R., & Novosad, P.** (2021). Development Research at High Geographic Resolution: An Analysis of Night-Lights, Firms, and Poverty in India using the SHRUG Open Data Platform. *The World Bank Economic Review*, 35(4). <https://doi.org/10.1093/wber/lhab003>
- Cai, W., Borlace, S., Lengaigne, M., Van Rensch, P., Collins, M., Vecchi, G., Timmermann, A., Santoso, A., McPhaden, M. J., Wu, L., & others** (2014). Increasing frequency of extreme El Niño events due to greenhouse warming. *Nature Climate Change*, 4(2), 111–116. <https://doi.org/10.1038/nclimate2100>
- Currie, J. & Almond, D.** (2011). Human capital development before age five. *Handbook of Labor Economics*, volume 4, 1315–1486. Elsevier. [https://doi.org/10.1016/S0169-7218\(11\)02413-0](https://doi.org/10.1016/S0169-7218(11)02413-0)
- Dixit, A.** (2009). Kosi embankment breach in Nepal: Need for a paradigm shift in responding to floods. *Economic and Political Weekly*, 70–78. <https://www.jstor.org/stable/40278487>
- Eriksson, J. G., Kajantie, E., Osmond, C., Thornburg, K., & Barker, D. J.** (2010). Boys live dangerously in the womb. *American Journal of Human Biology*, 22(3), 330–335. <https://doi.org/10.1002/ajhb.20995>
- Fagnäs, S. & Pelkonen, P.** (2020). Teachers, electoral cycles, and learning in India. *Journal of Human Resources*, 55(2), 699–732. <https://doi.org/10.3368/jhr.55.3.0816-8100R>
- Groppo, V. & Kraehnert, K.** (2016). Extreme weather events and child height: Evidence from Mongolia. *World Development*, 86, 59–78. <https://doi.org/10.1016/j.worlddev.2016.05.005>
- Heckman, J. J., Stixrud, J., & Urzua, S.** (2006). The effects of cognitive and noncognitive abilities on labor market outcomes and social behavior. *Journal of Labor Economics*, 24(3), 411–482. <https://doi.org/10.1086/504455>
- Henderson, J. V., Storeygard, A., & Weil, D. N.** (2011). A Bright Idea for Measuring Economic Growth. *American Economic Review*. <https://doi.org/10.1257/aer.101.3.194>
- Karbownik, K. & Wray, A.** (2019). Long-run consequences of exposure to natural disasters. *Journal of Labor Economics*, 37(3), 949–1007. <https://doi.org/10.1086/702652>
- Khanna, M. & Kochhar, N.** (2023). Do marriage markets respond to a natural disaster? The impact of flooding of the Kosi river in India. *Journal of Population Economics*, 36(4), 2241–2276. <https://doi.org/10.1007/s00148-023-00955-z>

- Kraemer, S.** (2000). The fragile male. *BMJ*, 321(7276), 1609–1612. <https://doi.org/10.1136/bmj.321.7276.1609>
- Krämer, M., Kumar, S., & Vollmer, S.** (2021). Improving child health and cognition: Evidence from a school-based nutrition intervention in India. *Review of Economics and Statistics*, 103(5), 818–834. https://doi.org/10.1162/rest_a_00950
- Lahoti, R. & Sahoo, S.** (2020). Are educated leaders good for education? Evidence from India. *Journal of Economic Behavior & Organization*, 176, 42–62. <https://doi.org/10.1016/j.jebo.2020.03.026>
- Leight, J., Glewwe, P., & Park, A.** (2015). The impact of early childhood rainfall shocks on the evolution of cognitive and non-cognitive skills.
- Lu, C., Liao, H., Wang, L., Li, Q., Liu, Q., & Wang, F.** (2023). The role of meteorological parameters on childhood asthma: Identifying critical windows of susceptibility during pregnancy. *Building and Environment*, 243, 110668. <https://doi.org/10.1016/j.buildenv.2023.110668>
- Pazos, N., Favara, M., Sánchez, A., Scott, D., & Behrman, J.** (2024). Long-term effects of early life rainfall shocks on foundational cognitive skills: Evidence from Peru. *Economics & Human Biology*, 54, 101407. <https://doi.org/10.1016/j.ehb.2024.101407>
- Rambachan, A. & Roth, J.** (2023). A more credible approach to parallel trends. *Review of Economic Studies*, 90(5), 2555–2591. <https://doi.org/10.1093/restud/rdad018>
- Rosales-Rueda, M.** (2018). The impact of early life shocks on human capital formation: Evidence from El Niño floods in Ecuador. *Journal of Health Economics*, 62, 13–44. <https://doi.org/10.1016/j.jhealeco.2018.07.003>
- Scholte, R. S., Van Den Berg, G. J., & Lindeboom, M.** (2015). Long-run effects of gestation during the dutch hunger winter famine on labor market and hospitalization outcomes. *Journal of Health Economics*, 39, 17–30. <https://doi.org/10.1016/j.jhealeco.2014.10.002>
- Shah, M. & Steinberg, B. M.** (2017). Drought of opportunities: Contemporaneous and long-term impacts of rainfall shocks on human capital. *Journal of Political Economy*, 125(2), 527–561. <https://doi.org/10.1086/690828>
- UNDP India** (2009). *Kosi floods 2008: how we coped! what we need? perception survey on impact and recovery strategies*. <https://www.undp.org/india/publications/kosi-floods-2008-how-we-coped-what-we-need-perception-survey-impact-and-recovery-strategies>

Webb, D. (2024). Critical Periods in Cognitive and Socioemotional Development: Evidence from Weather Shocks in Indonesia. *The Economic Journal*, 134(660), 1637–1665. <https://doi.org/10.1093/ej/uead105>

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Tables and Figures

Table 1: **Summary Statistics**

	Mean	SD	Min	Max	N
Reading Score	2.412	1.612	0	4	61,868
Math Score	2.334	1.455	0	4	61,868
Standardized Reading Score ^a	-0.025	1.005	-1.529	0.965	61,868
Standardized Math Score ^a	-0.022	1.006	-1.635	1.129	61,868
On Track (1=Yes)	0.810	0.393	0	1	61,868
Paid Private Tuition (1=Yes)	0.558	0.497	0	1	60,521
Dropout (1=Yes)	0.000	0.000	0	0	61,868
Never Enrolled (1=Yes)	0.000	0.000	0	0	61,868
Gender (1=Male)	0.498	0.500	0	1	61,868
Age	10.299	3.043	5	16	61,868
Class	4.871	2.870	1	12	61,868
Mother Literate (1=Yes)	0.364	0.481	0	1	61,868
Mother's Age	35.374	7.206	17	99	61,868
Household Size	7.513	3.576	1	78	61,868
Socioeconomic Status (1=High) ^b	0.645	0.478	0	1	61,868
Household Electricity (1=Yes)	0.449	0.497	0	1	61,868
Household Toilet (1=Yes)	0.234	0.423	0	1	61,868

Notes: Author's calculations from ASER household surveys 2013-14 (restricted to Bihar only).

^a The test score variables are standardized using the mean and standard deviation of the control group (i.e., children from unaffected districts) such that the mean and standard deviation of the standardized scores for them are 0 and 1, respectively.

^b Socioeconomic status (SES) is defined in absolute terms based on the type of dwelling. Children residing in houses made of concrete or semi-concrete material are categorized as higher SES (Non-poor), while those residing in thatched or mud houses are categorized as lower SES (Poor).

Table 2: **Main Results** — Effect of early-life exposure to flood on learning outcomes

	Standardized Reading Score			Standardized Math Score		
	(1)	(2)	(3)	(4)	(5)	(6)
Treated*Post	-0.079*** (0.028)	-0.078*** (0.028)	-0.079*** (0.029)	-0.059** (0.024)	-0.059** (0.024)	-0.061** (0.024)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	61868	61868	61868	61868	61868	61868
R-Squared	0.501	0.501	0.504	0.506	0.506	0.508
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No	Yes	Yes
District-Year FE	No	No	Yes	No	No	Yes

Notes: Treated takes value 1 for children from flood-affected districts and 0 otherwise; Post takes the value 1 for cohorts in utero or aged zero to five in 2008 and zero otherwise; regressions include controls for child's gender and grade, her mother's age and schooling status, socioeconomic status, household size, electricity and toilet availability; standard errors are reported in parentheses and are clustered at the district level. *, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively.

Data Source: ASER household surveys 2013-14 (restricted to Bihar only).

Table 3: **Heterogeneity** — Effects by gender, socioeconomic status and mother's literacy

	Standardized Reading Score			Standardized Math Score		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Heterogeneity by gender of the child</i>						
	Boys	Girls	DDD	Boys	Girls	DDD
Treated*Post	-0.093*** (0.032)	-0.067* (0.034)	-0.065* (0.034)	-0.054* (0.027)	-0.067** (0.028)	-0.065** (0.028)
Treated*Post*Boy			-0.029 (0.032)			0.007 (0.026)
Observations	30813	31055	61868	30813	31055	61868
R-Squared	0.506	0.505	0.505	0.513	0.505	0.509
<i>Panel B: Heterogeneity by socioeconomic status of the child</i>						
	High SES	Low SES	DDD	High SES	Low SES	DDD
Treated*Post	-0.028 (0.032)	-0.069** (0.030)	-0.071** (0.027)	-0.028 (0.025)	-0.053* (0.028)	-0.053* (0.027)
Treated*Post*High SES			0.042 (0.033)			0.025 (0.032)
Observations	39909	21959	61868	39909	21959	61868
R-Squared	0.489	0.509	0.505	0.495	0.507	0.509
<i>Panel C: Heterogeneity by mother's literacy</i>						
	Literate Mom	Illiterate Mom	DDD	Literate Mom	Illiterate Mom	DDD
Treated*Post	-0.096*** (0.027)	-0.058* (0.033)	-0.055* (0.031)	-0.060** (0.025)	-0.053* (0.030)	-0.048 (0.029)
Treated*Post*Literate Mom			-0.052* (0.028)			-0.025 (0.032)
Observations	22508	39360	61868	22508	39360	61868
R-Squared	0.479	0.502	0.508	0.490	0.500	0.510
Controls	Yes	Yes	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
District-Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: See 2 notes; *, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively

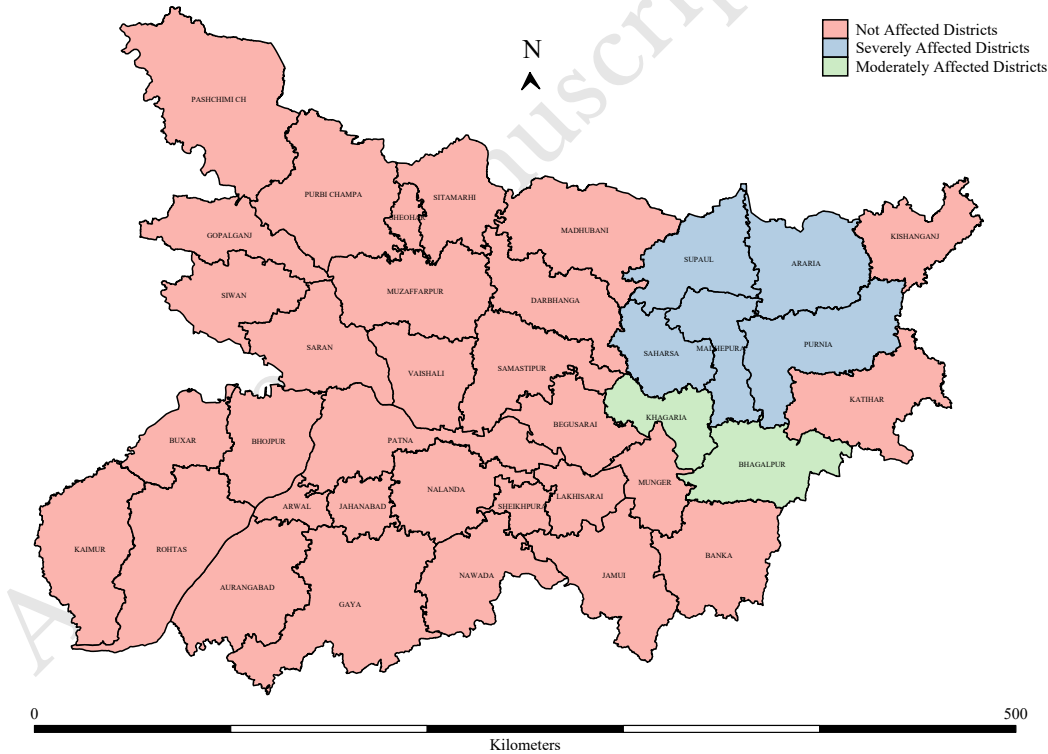
Table 4: **Mechanism** — Effect of early-life exposure to flood on nutrition

	Height-for-age (<i>z-score</i>)			
	(1)	(2)	(3)	(4)
Treated*Post	0.182 (0.121)	0.172 (0.116)	0.182 (0.121)	0.159 (0.116)
Controls	No	No	No	Yes
Observations	2171	2171	2171	2171
R-Squared	0.071	0.061	0.071	0.107
District FE	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes
Religion FE	No	Yes	Yes	Yes
Caste FE	No	No	Yes	Yes

Notes: Treated takes the value 1 for children from flood-affected districts and 0 otherwise; Post takes the value 1 for cohorts in utero or aged zero to five in 2008 and zero otherwise; regressions include controls for child's gender, dummy for urban, household size, per-capita consumption, households assets and highest adult education within the household; standard errors are reported in parentheses and are clustered at the district level. *, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively.

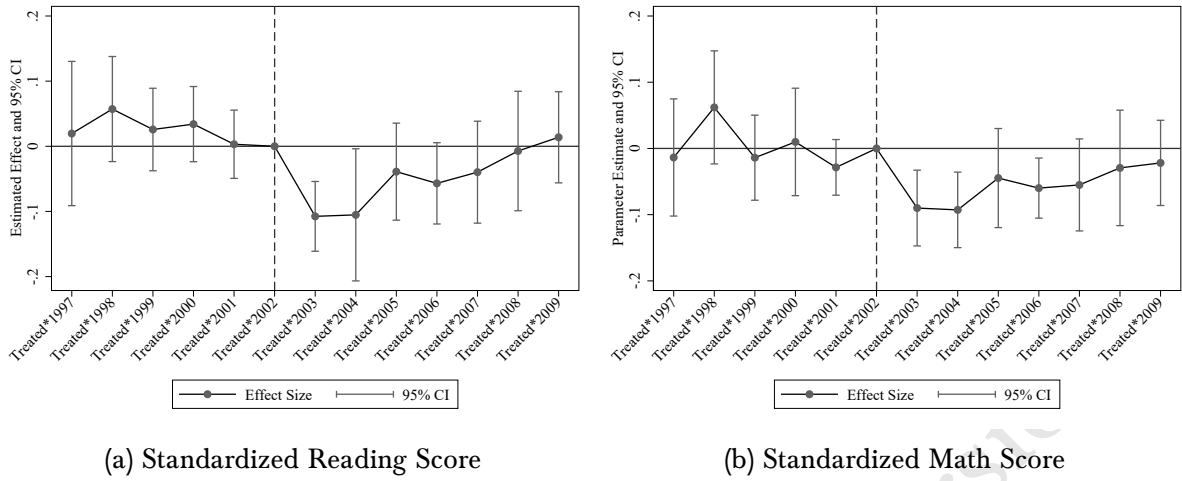
Data Source: IHDS-II individual-level data 2011-12 (restricted to Bihar only)

Figure 1: Districts affected by 'Kosi' flood in August 2008



Notes: District Map of Bihar, showing the districts affected and not affected by the 'Kosi flood' of 2008

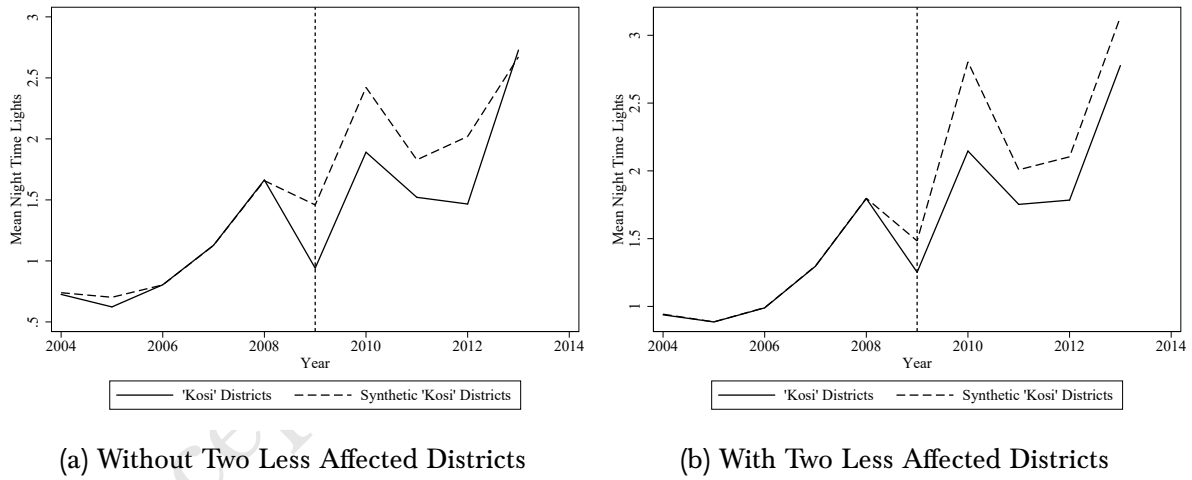
Figure 2: **Event Study** — Cohort-wise effects of exposure to flood on learning outcomes



Notes: The figure reports the event study estimates from estimating equation 2; *Treated* refers to the districts affected by the *Kosi* flood; the years in the figure are birth years of children and refer to birth cohorts; the dots represent the estimated coefficients and the bars are the associated 95% confidence intervals

Data Source: ASER household surveys 2013-14 (restricted to Bihar only).

Figure 3: **Mechanism** — Effects of *Kosi* flood on economic activity at the district level



Notes: The figure uses Night Time Lights and reports the outcome paths for *Kosi* districts and synthetic *Kosi* districts; the synthetic version is constructed from a bunch of non-*Kosi* districts using the synthetic control method; we match based on the average night-time lights in the pre-period.

Data Source: Night Time Lights data (2004-2013) extracted from the SHRUG database (Henderson et al., 2011; Asher et al., 2021).

EARLY-LIFE EXPOSURE TO WEATHER SHOCKS AND LEARNING OUTCOMES IN INDIA

Ather Hassan Dar

Appendix Material
[Journal of Environmental Management]

1. Additional Tables

Table A.1: **Main Results** — Effect of early-life exposure to flood on learning outcomes:
Extended Version

	(1)	(2)	(3)	(4)	(5)
<i>Panel A: Standardized Reading Score</i>					
Treated*Post	-0.108*** (0.033)	-0.107*** (0.032)	-0.079*** (0.028)	-0.078*** (0.028)	-0.079*** (0.029)
Controls	Yes	Yes	Yes	Yes	Yes
Observations	61868	61868	61868	61868	61868
R-Squared	0.460	0.466	0.501	0.501	0.504
	(6)	(7)	(8)	(9)	(10)
<i>Panel B: Standardized Math Score</i>					
Treated*Post	-0.087*** (0.028)	-0.088*** (0.027)	-0.059** (0.024)	-0.059** (0.024)	-0.061** (0.024)
Controls	Yes	Yes	Yes	Yes	Yes
Observations	61868	61868	61868	61868	61868
R-Squared	0.463	0.470	0.506	0.506	0.508
District FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	No	Yes	Yes	Yes
Year FE	No	No	No	Yes	Yes
District-Year FE	No	No	No	No	Yes

Note: Treated takes value 1 for children from flood-affected districts and 0 otherwise; Post takes the value 1 for cohorts in utero or aged zero to five in 2008 and zero otherwise; regressions include controls for child's gender and grade, her mother's age and schooling status, socioeconomic status, household size, electricity and toilet availability; standard errors are reported in parentheses and are clustered at the district level. *, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively.

Data Source: ASER household surveys 2013-14 (restricted to Bihar only).

Table A.2: **Additional Results** — Effect of early-life exposure to flood on the probability of being on track and uptake of private tuition

	On Track (1=Yes)			Private Tuition (1=Yes)		
	(1)	(2)	(3)	(4)	(5)	(6)
Treated*Post	-0.011 (0.010)	-0.014 (0.010)	-0.015 (0.010)	0.013 (0.009)	0.013 (0.009)	0.010 (0.010)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	61868	61868	61868	60521	60521	60521
R-Squared	0.504	0.545	0.546	0.108	0.108	0.111
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No	Yes	Yes
District-Year FE	No	No	Yes	No	No	Yes

Note: See Table A.1 notes; 'On Track' takes a value of 1 if a child is enrolled in a grade appropriate for their age (with a typical school-start age in India being 6 years); *, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively.

Table A.3: **Main Results** — Effect of early-life exposure to flood on learning outcomes in the short-run

	Standardized Reading Score			Standardized Math Score		
	(1)	(2)	(3)	(4)	(5)	(6)
Treated*Post	0.032 (0.043)	0.032 (0.042)	0.023 (0.042)	0.010 (0.045)	0.009 (0.045)	0.002 (0.044)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	74842	74842	74842	74842	74842	74842
R-Squared	0.435	0.435	0.445	0.443	0.443	0.455
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No	Yes	Yes
District-Year FE	No	No	Yes	No	No	Yes

Note: See Table A.1 notes; *, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively.

Data Source: ASER household surveys 2009-10 (restricted to Bihar only).

Table A.4: **Robustness** — Main results without less affected districts

	Standardized Reading Score			Standardized Math Score		
	(1)	(2)	(3)	(4)	(5)	(6)
Treated*Post	-0.111*** (0.023)	-0.110*** (0.023)	-0.110*** (0.023)	-0.090*** (0.015)	-0.090*** (0.015)	-0.092*** (0.015)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	58615	58615	58615	58615	58615	58615
R-Squared	0.501	0.501	0.504	0.506	0.506	0.508
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No	Yes	Yes
District-Year FE	No	No	Yes	No	No	Yes

Note: See Table A.1 notes; *, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively.

Table A.5: **Robustness** — Main results without neighboring districts

	Standardized Reading Score			Standardized Math Score		
	(1)	(2)	(3)	(4)	(5)	(6)
Treated*Post	-0.087*** (0.029)	-0.086*** (0.029)	-0.087*** (0.029)	-0.069** (0.026)	-0.069** (0.026)	-0.072** (0.026)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	49109	49109	49109	49109	49109	49109
R-Squared	0.502	0.502	0.504	0.504	0.504	0.505
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No	Yes	Yes
District-Year FE	No	No	Yes	No	No	Yes

Note: See Table A.1 notes; *, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively.

Table A.6: **Robustness** — Main results with Logit models

	A child can read		A child can solve	
	Paragraph	Story	Subtraction	Division
	(1)	(2)	(3)	(4)
Treated*Post	-0.156** (0.079)	-0.172** (0.075)	-0.096* (0.056)	-0.154** (0.076)
Controls	Yes	Yes	Yes	Yes
Observations	61868	61868	61868	61868
Pseudo R-Squared	0.378	0.352	0.340	0.307
District FE	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
District-Year FE	Yes	Yes	Yes	Yes

Note: See Table A.1 notes; the dependent variables are dummies that take a value of 1 if a child can read and solve arithmetic problems as indicated by model labels and 0 otherwise; *, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively.

Table A.7: **Robustness** — Main results without outliers

	Standardized Reading Score			Standardized Math Score		
	(1)	(2)	(3)	(4)	(5)	(6)
Treated*Post	-0.078*** (0.029)	-0.078** (0.029)	-0.078** (0.029)	-0.062** (0.023)	-0.062** (0.023)	-0.064*** (0.023)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	60710	60710	60710	60710	60710	60710
R-Squared	0.502	0.502	0.505	0.506	0.506	0.509
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No	Yes	Yes
District-Year FE	No	No	Yes	No	No	Yes

Note: See Table A.1 notes; *, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively.

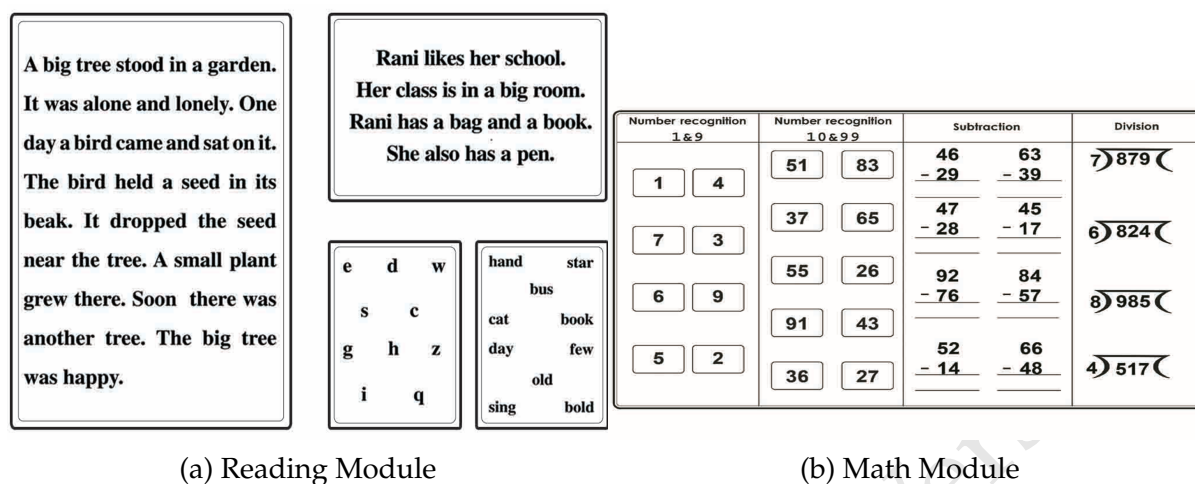
Table A.8: **Robustness** — Effect of early-life exposure to flood on learning outcomes in Uttarakhand

	Standardized Reading Score			Standardized Math Score		
	(1)	(2)	(3)	(4)	(5)	(6)
Treated*Post	-0.407*** (0.071)	-0.404*** (0.073)	-0.403*** (0.072)	-0.373*** (0.082)	-0.374*** (0.082)	-0.372*** (0.084)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16743	16743	16743	16743	16743	16743
R-Squared	0.426	0.426	0.428	0.404	0.404	0.406
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No	Yes	Yes
District-Year FE	No	No	Yes	No	No	Yes

Note: Treated takes value 1 for children from flood-affected districts in Uttarakhand and 0 for children in Himachal Pradesh; Post takes the value 1 for cohorts in utero or aged zero to five in 2013 and zero otherwise; regressions include controls for child's gender and grade, her mother's age and schooling status, socioeconomic status, household size, electricity and toilet availability; standard errors are reported in parentheses and are clustered at the district level. *, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively.

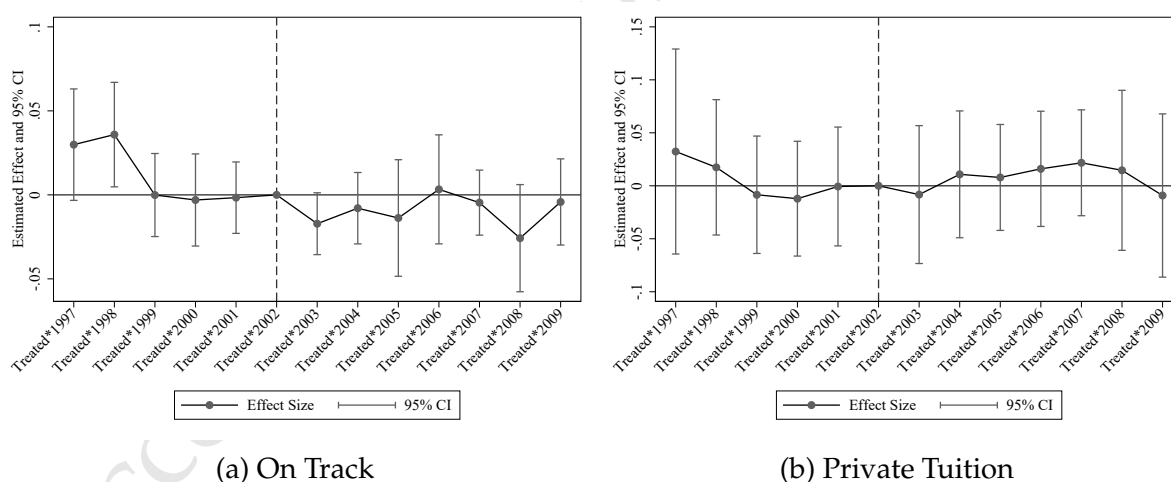
Data Source: ASER household surveys 2017-18 (restricted to Uttarakhand and Himachal Pradesh only).

Figure A.1: **ASER test tools** — A sample of the reading and math modules of ASER Surveys



2. Additional Figures

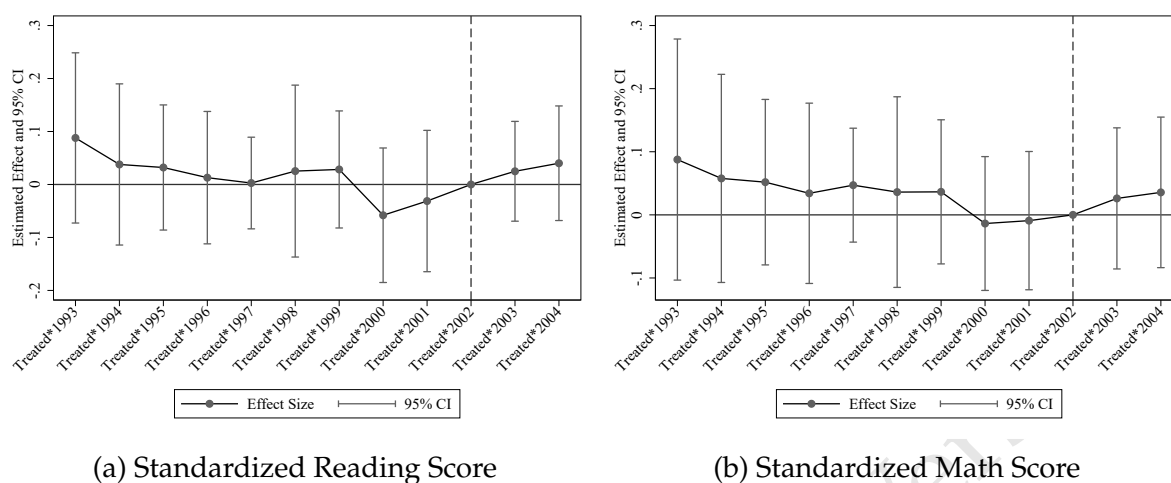
Figure A.2: **Event Study** — Cohort-wise effects of exposure to flood on the probability of being on track and uptake of private tuition



Note: The figure reports the event study estimates from estimating equation 2; *Treated* refers to the districts affected by the *Kosi* flood; the years in the figure are birth years of children and refer to birth cohorts; the dots represent the estimated coefficients and the bars are the associated 95% confidence intervals

Data Source: ASER household surveys 2013-14 (restricted to Bihar only).

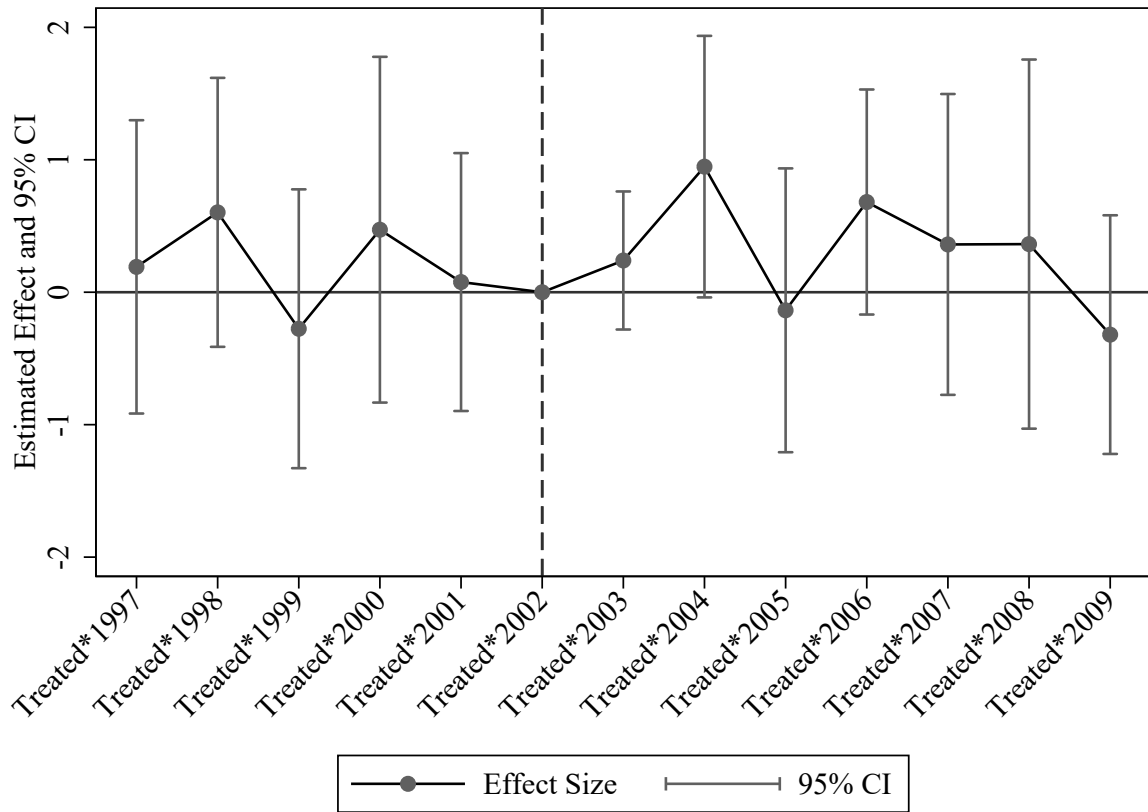
Figure A.3: **Event Study** — Cohort-wise effects of exposure to flood on learning outcomes in the short-run



Note: The figure reports the event study estimates from estimating equation 2; *Treated* refers to the districts affected by the *Kosi* flood; the years in the figure are birth years of children and refer to birth cohorts; the dots represent the estimated coefficients and the bars are the associated 95% confidence intervals

Data Source: ASER household surveys 2009-10 (restricted to Bihar only).

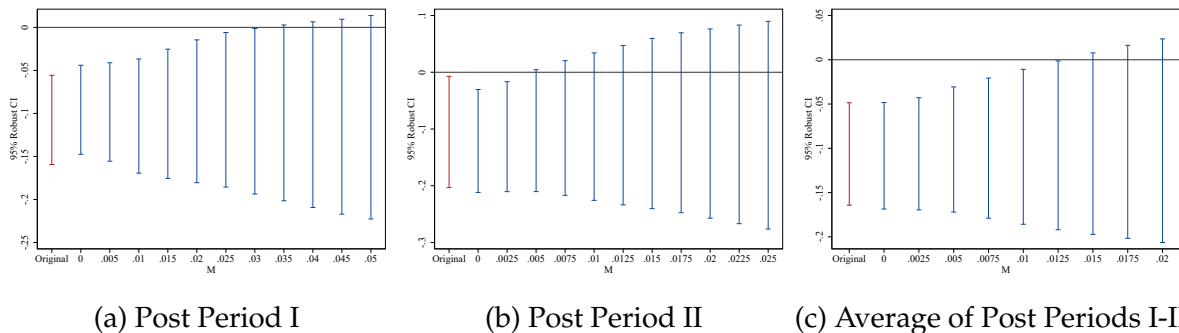
Figure A.4: **Event Study** — Cohort-wise effects of exposure to flood on height-for-age (z-score)



Note: The figure reports the event study estimates for height-for-age; *Treated* refers to the districts affected by the *Kosi* flood; the years in the figure are birth years of children and refer to birth cohorts; the dots represent the estimated coefficients and the bars are the associated 95% confidence intervals

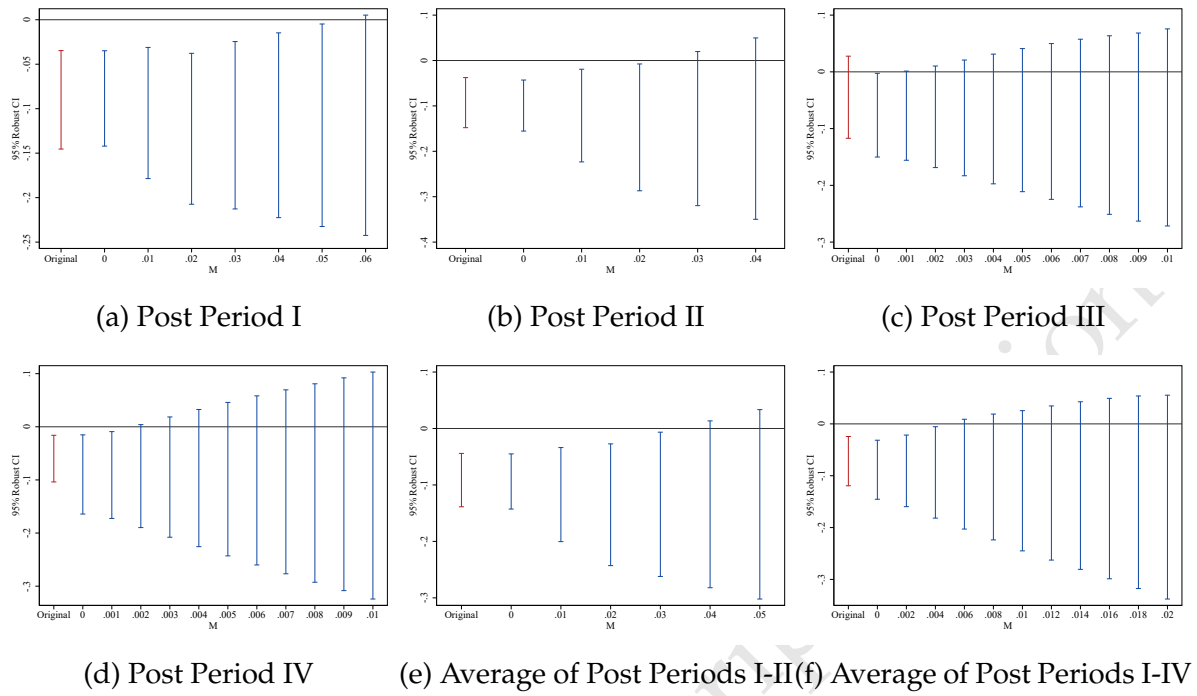
Data Source: IHDS-II individual-level data 2011-12 (restricted to Bihar only).

Figure A.5: **Sensitivity Analysis** — Testing the sensitivity of estimated effects to the potential violation of parallel trends (Standardized Reading Score)



Note: Accounting for potential violations of the parallel trends assumption, this figure presents a sensitivity analysis of the estimated treatment effects in each post-period; The red bars indicate the original 95% confidence intervals, while the blue bars depict the robust 95% confidence intervals under the assumption that parallel trends may deviate by up to M times per period from the observed pre-trend; Here, M represents the maximum allowable divergence in the underlying trend slope between consecutive time periods.

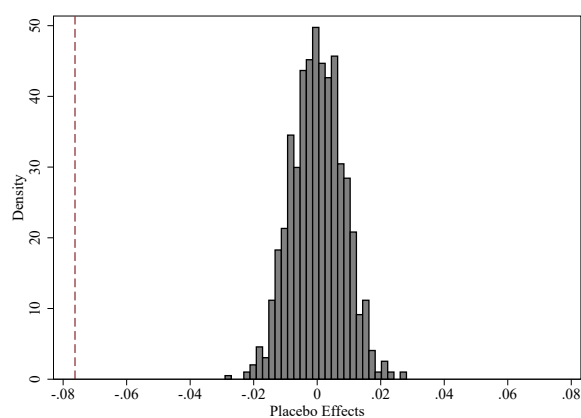
Figure A.6: **Sensitivity Analysis** — Testing the sensitivity of estimated effects to the potential violation of parallel trends (Standardized Math Score)



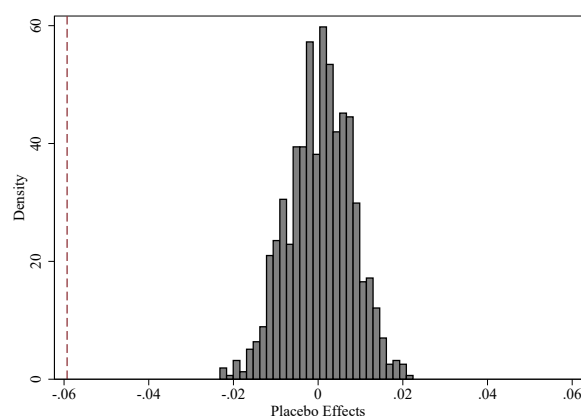
Note: See Figure A.5 notes.

Figure A.7: **Randomization** — Placebo effects of early-life exposure to flood on learning outcomes

I. Random Assignment of Districts to Treatment

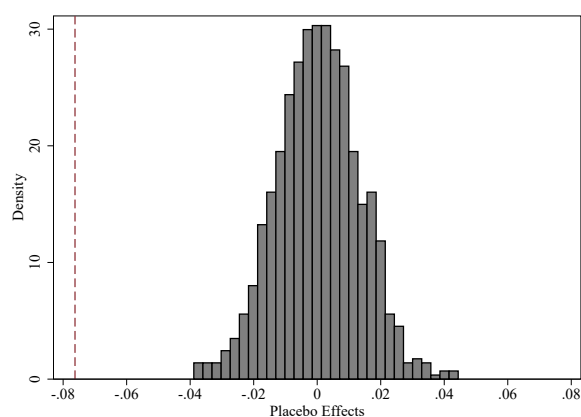


(a) Standardized Reading Score

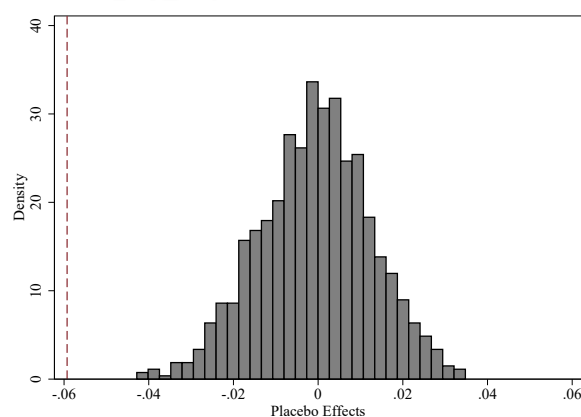


(b) Standardized Math Score

II. Random Assignment of Cohorts to Treatment



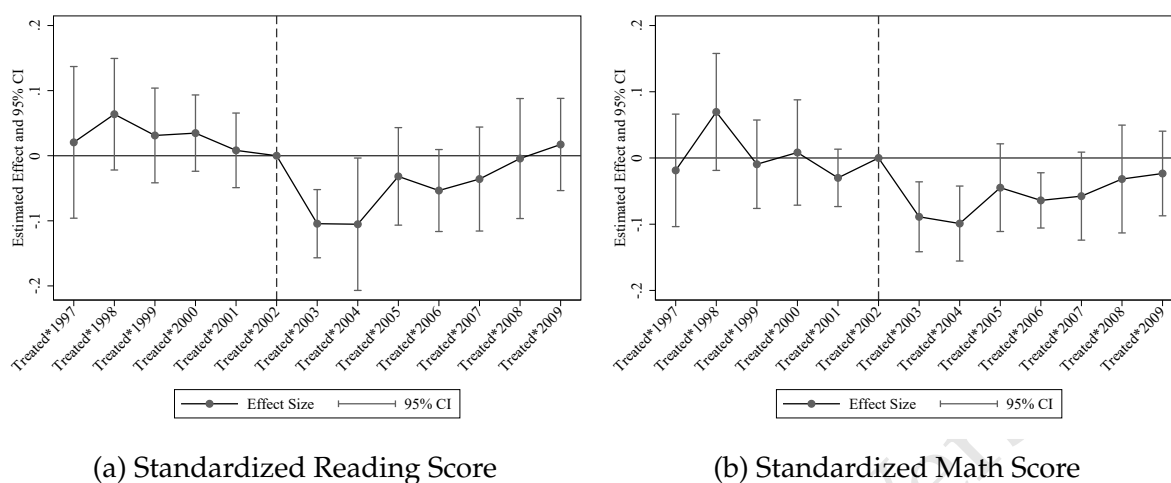
(c) Standardized Reading Score



(d) Standardized Math Score

Note: We conduct the randomization exercise using our baseline model (1), where we randomly assign treatment status to districts and cohorts 1,000 times; for each iteration, we re-estimate the model and obtain pseudo-treatment effects; we repeat the process 1,000 times and compile the coefficient estimates; we plot the distribution of these effects in this figure; the original estimate is highlighted with a red dashed line.

Figure A.8: **Event Study** — Cohort-wise effects of exposure to flood on learning outcomes without outliers



Note: The figure reports the event study estimates from estimating equation 2; *Treated* refers to the districts affected by the *Kosi* flood; the years in the figure are birth years of children and refer to birth cohorts; the dots represent the estimated coefficients and the bars are the associated 95% confidence intervals

Data Source: ASER household surveys 2013-14 (restricted to Bihar only).